

Ques Given the three integers n , a and b . Return the n^{th} magical number. As the result can be large, return $\% 10^9 + 7$.
A positive integer is magical no. if it is divisibly by a & b .

Ans

```
public int gcd(int a, int b) {
    if (b == 0) {
        return a;
    }
    return gcd(b, a % b);
}
```

```
public int nthMagicalNumber(int n, int a, int b) {
    long long gcd = gcd(a, b);
    long lcm = (a * b) / gcd;
    long start = Math.min(a, b), end = Math.min(a, b) * n;
    long ans = 0;
    while (start <= end) {
        long count =
        long mid = start + (end - start) / 2;
        long count = mid / a + mid / b - mid / lcm;
        if (count == n) {
            ans = mid;
            end = mid - 1;
        } else if (count < n) {
            start = mid + 1;
        } else {
            end = mid - 1;
        }
    }
    int MOD = (int) 1e9 + 7;
    return (int) (ans % MOD);
}
```

Time Complexity $\rightarrow$
 $(\log(\min(a, b)) + \log(\min(a, b) * n))$
 $\downarrow \qquad \qquad \downarrow$
gcd function

Space Complexity $\rightarrow$
 $(\log(\min(a, b)))$
 $\downarrow$
gcd recursion depth